



# **BRAC University**

Department of Mathematics and Natural Sciences

**Total Points: 150**

 **Assignment - 01**

**Course Code: MAT 215**

Complex Variables and Laplace Transformations

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 **Section: 12**

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 **Submission Date: \_\_\_\_\_**

*Assigned by*

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### Question 1

Find all possible values of  $z$  satisfying

$$z^6 = -\frac{729}{2} + \frac{729\sqrt{3}i}{2}.$$

Locate them on the complex plane. Show that they lie on a circle, and determine its radius. Also, find the angular distance between two adjacent roots.

 **Solution:**

## Question 2

Describe the locus  $|z - 6i| - |z + 6i| = 7$  on the complex plane.

 Solution:

### Question 3

Describe the region  $|z - 8i| - |z + 8i| > 12$  on the complex plane.

 Solution:

### Question 4

Solve the equation

$$e^{2z} = -2\sqrt{3} - 2i$$

for  $z$  and express  $z$  as  $x + iy$  where  $x, y \in \mathbb{R}$ .

 **Solution:**

### Question 5

Prove that

$$\sec^{-1} z = \frac{1}{i} \ln \left( \frac{1 + \sqrt{1 - z^2}}{z} \right).$$

 Solution:

### Question 6

Solve for  $z$  where

$$\operatorname{cosech} z = 7 + 8i.$$

 Solution:

### Question 7

Using the definition of a limit, show that  $\lim_{z \rightarrow 0} \frac{\operatorname{Re}(z^2)}{|z|^2}$  does not exist.

 Solution:

## Question 8

Using L'Hôpital's rule, evaluate

$$\lim_{z \rightarrow 0} \left( \frac{\sin z}{z} \right)^{\frac{9 \sin(3z)}{z - \sin z}}.$$

 Solution:

### Question 9

Consider the function

$$f(z) = \frac{\tan 4z}{9z}.$$

Is  $f(z)$  continuous at  $z = 0$ ? If not, redefine  $f$  at  $z = 0$  so that  $f(z)$  becomes continuous. Also, find all points of discontinuity of  $f(z)$ .

 **Solution:**

### Question 10

Using the definition, show that

$$f(z) = 5z^2 + 2z - 4$$

is differentiable at all points. Also find the derivative.

 **Solution:**

### Question 11

Using the definition, find the derivative of  $f(z) = \frac{6}{z^2}$  at  $z = 9 + 5i$ .

 Solution:

## Question 12

Consider the function

$$f(z) = 5 \sin(4z) - 8 \cosh(3z).$$

Using the Cauchy–Riemann equations, determine whether the function is analytic.

 **Solution:**

### Question 13

Consider the function

$$f(z) = 7ze^{-9z}.$$

Using the Cauchy–Riemann equations, determine whether the function is analytic.

 **Solution:**

### Question 14

Show that the function

$$V(x, y) = 8 \sin(4x) \cosh(4y) + 27x^2y - 2x^2 - 9y^3 + 2y^2$$

is harmonic. Find the harmonic conjugate  $U$  of  $V$  such that  $U + Vi$  becomes analytic.

 **Solution:**

### Question 15

Show that the function

$$V(x, y) = 4xe^{-5x} \cos(5y) + 4ye^{-5x} \sin(5y)$$

is harmonic. Find the harmonic conjugate  $U$  of  $V$  such that  $U + Vi$  becomes analytic.

 **Solution:**